

Computer Circuit

Condensed Study Notes

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An Introduction to Computer Circuit

Computer circuit speed is key to performance. Faster operation is achieved by speeding up internal switches and designing routes for high-speed electron flow.

A computer circuit is a complete path or network for electron flow, operating on a binary system with two states (on/off).

These circuits use transistors as electronic switches that open and close incredibly fast, in nanoseconds and picoseconds.

In theoretical computer science, a circuit is a computational model where input values pass through gates, each performing a function. This generalizes Boolean circuits and models digital logic.

Key Terms:

- **Circuit:** A complete path or network for electron flow in a computer.
- **Transistor:** An electronic switch that can be opened or closed rapidly.
- **Boolean Circuit:** A type of theoretical circuit using Boolean values and gates like conjunction, disjunction, and negation.

Motherboard as the main circuit board in a PC

The motherboard is the main printed circuit board (PCB) in a computer. It serves as the central communications backbone, connecting all internal components and external peripherals.

Key components connected through the motherboard include:

- Chipsets
- Central Processing Units (CPU)
- Memory

External peripherals that connect via the motherboard include:

- Wi-Fi
- Ethernet
- Graphics cards (with a Graphics Processing Unit, or GPU)

IC as the major Circuitry Technology in Modern PC

Integrated circuits (ICs), also known as microchips, are fundamental to modern electronics. These compact devices house thousands of transistors, resistors, and other components, enabling complex functions efficiently.

ICs come in various forms, sizes, and specifications to meet diverse needs. Examples include memory chips and microcontrollers found in automobiles.

Key components of an IC include transistors, resistors, and condensers, with multiple terminals for different functions. They are miniature and can be manufactured on a silicon wafer.

What is an Integrated Circuit

An Integrated Circuit (IC) is a compact electronic device containing multiple electronic components (like transistors) that are all connected and function as a single unit. These components are built on a single, flat piece of semiconductor material.

IC technology allows for circuits that are faster, more reliable, more efficient, and less expensive than traditional, discrete component circuits.

Based on their construction, ICs are divided into three types:

- a) Digital Integrated Circuit
- b) Analog Integrated Circuit
- c) Mixed Integrated Circuit